

### D003 human-response layer (empirical)

### Vehicle layer

### Automation-transition layer

### Scenario and human timing (model inputs)

t\_button  
(TOR → mode switch)  
[observed]

drag, t2, a\_max,  
friction  $a \leq \mu g$   
[source-supported]

P0 / P1 / P2  
counterfactual branches  
[model condition]

speed; TTC 2 and 7 s;  
 $\mu = 1.0$   
[source-supported]

exposure and covariate  
associations  
[observed]

t3 brake build-up  
[assumption]

authority timing;  
support a\_sup  
[assumption]

$\mu = 0.5$ ;  
TTC 3, 5, 10 s  
[assumption]

lane crossing;  
hazard distance  
[observed]

braking command  $u = 1$   
[model condition]

t1 effective-braking  
onset interval  
[assumption]

steering onset,  
indicator, gaps  
[provisional]

driver brake onset  
and strength  
[blocked]

× no arrow: no D003 human quantity enters the model (t\_button ≠ t1; no identified human → braking link)

R3B braking-only model: interval bounds on  $T_{\text{required}}^{\text{brake}}$   
labels ROBUSTLY SATISFIED / PARAMETER-SENSITIVE / ROBUSTLY UNSATISFIED  
[model condition]

□ observed    □ source-supported    □ assumption    □ model condition    □ provisional    ■ blocked